19 P has coordinates (–9, 7)
Q has coordinates (11, 12)

M is the point on the line segment PQ such that $PM : MQ = 2 : 3$

Line **L** is perpendicular to the line segment PQ.
L passes through M.

Find an equation of **L**.

Gradient of PQ = (12 – 7) ÷ (11 – (–9))

= 5 ÷ 20 = ¼.

1 mark
Find gradient of PQ

P to Q changes x by 20 and y by 5.

M is $\frac{2}{5}$ of the way from P to Q:

$x = -9 + \frac{2}{5} \times 20 = -1$

$y = 7 + \frac{2}{5} \times 5 = 9$, so M = (–1, 9).

1 mark
Find point M

Perpendicular gradient = –1 ÷ ¼ = –4.

1 mark
Negative reciprocal

Use $y = -4x + c$ and M = (–1, 9).

$$\begin{array}{ccc} 9 & = & 4 + c \\ \text{(-4)} & & \text{(-4)} \\ 5 & = & c \end{array}$$

1 mark
Substitute M and m

Line L: $y = -4x + 5$.

1 mark
y = –4x + 5

(Total for Question 19 is 5 marks)